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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I Kothakota Rakesh/192525075 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Total						

QUESTIONS:01

Total Marks:25

Select and review a recent research article related to the below **topic**, such as **Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM**. Summarize the **problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings**, and critically comment on the **advantages, limitations, and possible future improvements** of the approach presented in the article.

Prepare a structured Article Review addressing the following sections:

1. Article Details

Provide the following information:

Title of the research article

Names of the authors

Journal/Conference name

Year of publication

DOI or publication link

Topic/Deep Learning technique addressed

Title: *Long Short-Term Memory (LSTM) Based Deep Learning Models for Predicting Univariate Time Series Data*

Authors: Ashish Kharel, Zeinab Zarean, and Devinder Kaur

Journal: International Journal of Machine Learning (IJML)

Year of Publication: 2024

Volume and Pages: Volume 14, No. 1, pp. 30–37

DOI: 10.18178/ijml.2024.14.1.1154

Topic: Long Short-Term Memory (LSTM), Deep Learning, and Time-Series Prediction
The article compares three LSTM-based models—**Vanilla LSTM, Bidirectional LSTM, and Stacked LSTM**—for predicting COVID-19 case trends.

2. Research Problem

What problem does the article attempt to solve?

Why is the problem important?

What are the limitations of existing approaches identified by the authors?

The research addresses the problem of **predicting the number of COVID-19 cases using historical time-series data**.

Accurate prediction of COVID-19 trends is important because it can help governments and healthcare organizations prepare medical resources, healthcare professionals, medicines, and other infrastructure.

Traditional prediction approaches can have difficulty capturing complex temporal relationships in sequential data. Therefore, the authors investigate LSTM-based deep learning models that can learn patterns from previous observations.

The main research problem is:

How effectively can different LSTM architectures predict future COVID-19 cases from historical daily case data?

3. Objectives of the Study

What are the main objectives of the research?

What specific goals does the proposed model attempt to achieve?

The main objectives are:

1. To investigate LSTM models for univariate time-series prediction.
2. To predict COVID-19 case trends in the United States.
3. To compare different LSTM architectures.
4. To study the performance of **Vanilla LSTM, Bidirectional LSTM, and Stacked LSTM**.
5. To determine which LSTM architecture provides the best prediction accuracy.
6. To optimize important LSTM hyperparameters.

The study ultimately found that **Bidirectional LSTM provided the best accuracy among the three investigated models**.

4. Dataset Description

What dataset(s) are used?

What is the size of the dataset?

What type of data is used—images, text, time-series, video, etc.?

How is the dataset divided into training, validation, and testing sets?

The study uses **daily COVID-19 case data from the United States**.

Dataset source: Centers for Disease Control and Prevention (CDC)

Data type: Univariate time-series data

Period: January 23, 2020 – May 25, 2022

Total samples: 847

Training data: 743 samples

Testing data: 104 samples

The data represents the daily trend in the number of COVID-19 cases.

The authors trained the models using the first 743 observations and evaluated them on the final 104 observations.

Dataset Flow

COVID-19 Daily Cases → Preprocessing → Training Data (743) → LSTM Model → Testing Data (104) → Prediction

5. Proposed Methodology

Explain the overall methodology adopted in the article.

Identify the deep learning architecture used.

Explain the important layers/components of the model.

Describe the data preprocessing techniques used.

Explain the training procedure and important parameters, wherever available.

The overall methodology consists of the following steps:

Step 1 – Data Collection

COVID-19 daily case data was collected from the CDC.

Step 2 – Data Preparation

The collected data was organized as a univariate time series suitable for sequential prediction.

Step 3 – Dataset Division

The first **743 samples** were used for training and the final **104 samples** were used for testing.

Step 4 – Model Development

Three different LSTM architectures were developed:

- Vanilla LSTM
- Bidirectional LSTM
- Stacked LSTM
- **Step 5 – Hyperparameter Optimization**
- The models were implemented using **TensorFlow**, while **KerasTuner** was used to optimize LSTM hyperparameters.
- **Step 6 – Model Training**
- The models learned temporal relationships from historical COVID-19 case observations.
- **Step 7 – Prediction**
- The trained models were used to predict future values.
- **Step 8 – Evaluation**
- The prediction performance was compared using **Mean Absolute Percentage Error (MAPE)**.

6. Deep Learning Technique Used

Explain how the selected technique is applied in the research.

How is Transfer Learning used?

How does DenseNet perform feature reuse?

How is RNN/LSTM used to process sequential data?

How does a Bidirectional RNN utilize information from both directions?

How does an Encoder–Decoder/Seq2Seq architecture process input and output sequences?

How is BPTT used for training recurrent networks?

Long Short-Term Memory (LSTM) is a special type of Recurrent Neural Network designed to process sequential data.

An LSTM contains three major gates:

1. Forget Gate

Determines which previous information should be discarded.

2. Input Gate

Determines which new information should be stored.

3. Output Gate

Determines which information should be passed to the next stage.

A simplified LSTM operation is:

Input → Forget Gate → Input Gate → Memory Cell → Output Gate → Prediction

LSTM is suitable for this research because COVID-19 case data is sequential and depends on patterns occurring over time.

Bidirectional LSTM

Bidirectional LSTM processes sequence information in two directions:

Forward direction →

← Backward direction

This allows the model to use information from both directions while learning the sequence representation.

The article compares Vanilla LSTM, Bidirectional LSTM, and Stacked LSTM.

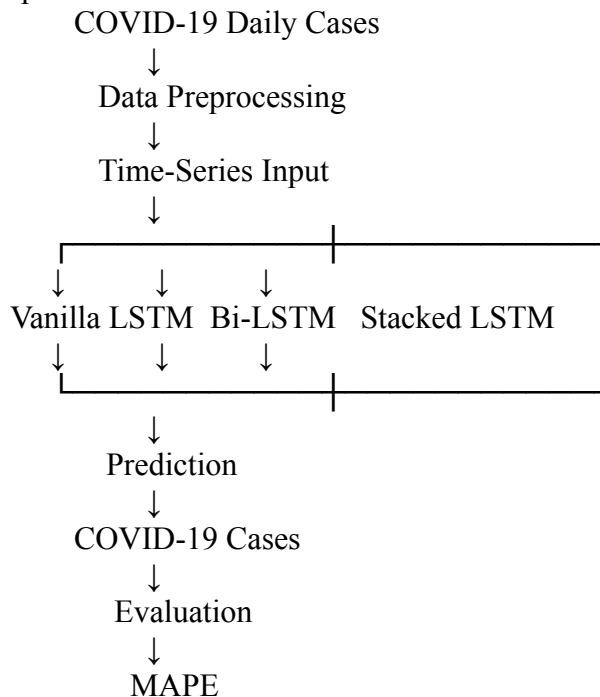
7. Model Architecture

Draw or reproduce a simplified architecture diagram of the proposed model and explain:

Input → Preprocessing → Feature Extraction/Encoder → Hidden Layers →
Classification/Decoder → Output

Identify the role of the major components in the architecture.

Simplified Architecture



Major Components

Input: Historical COVID-19 case values.

Preprocessing: Converts the collected data into a suitable sequential format.

LSTM Layer: Learns temporal dependencies.

Bidirectional LSTM: Processes information in both directions.

Stacked LSTM: Uses multiple LSTM layers to learn deeper representations.

Output: Predicted COVID-19 case values.

Evaluation: MAPE is used to compare prediction accuracy.

8. Performance Evaluation

Identify the evaluation metrics used in the article, such as:

Accuracy, Precision, Recall, F1-Score, Loss, Mean Squared Error (MSE)

BLEU Score, where applicable

Report the major performance results obtained by the proposed model.

The main evaluation metric used in the article is:

Mean Absolute Percentage Error (MAPE)

MAPE measures the average percentage difference between actual and predicted values.

The general formula is:

$$\text{MAPE} = (100/n) \times \sum |(\text{Actual} - \text{Predicted}) / \text{Actual}|$$

A lower MAPE indicates better prediction performance.

The researchers compared the three LSTM architectures using MAPE and reported that

Bidirectional LSTM achieved the best accuracy.

9. Results and Discussion

What are the major findings of the research?

How does the proposed model perform compared with existing methods?

Which model/technique produced the best results?
What factors contributed to the observed performance?

The major finding of the study is that different LSTM architectures produce different prediction performance.

The three models evaluated were:

Model	Purpose
Vanilla LSTM	Basic LSTM prediction
Bidirectional LSTM	Uses information in both directions
Stacked LSTM	Uses multiple LSTM layers

Best Performing Model

Bidirectional LSTM

The article reports that the **Bidirectional LSTM gave the best accuracy** among the three investigated models.

Discussion

The improved performance of Bidirectional LSTM can be attributed to its ability to process sequence information from both directions. This provides richer contextual information for learning patterns in the time series.

The use of KerasTuner for hyperparameter optimization also helped identify suitable model configurations.

10. Advantages of the Proposed Model

Identify at least three advantages of the proposed approach.

Advantage 1 – Handles Sequential Data

LSTM is specifically designed for sequential and time-series data.

Advantage 2 – Learns Long-Term Dependencies

LSTM can preserve useful information across multiple time steps.

Advantage 3 – Multiple Architectures Can Be Compared

The study evaluates Vanilla LSTM, Bidirectional LSTM, and Stacked LSTM rather than using only one architecture.

Advantage 4 – Hyperparameter Optimization

KerasTuner was used to optimize important model parameters.

Advantage 5 – Practical Application

The model is applied to a real-world COVID-19 dataset rather than a purely artificial dataset.

11. Limitations of the Study

Identify the major limitations mentioned by the authors and any additional limitations you observe, such as: Small dataset, High computational requirements, Training time, Overfitting, Limited generalization, Model complexity.

The following limitations can be identified:

1. The study uses **univariate data**, so only one primary variable is considered.
2. The model focuses on historical COVID-19 cases and does not directly incorporate other factors such as vaccination, mobility, weather, or government policies.
3. The dataset is specific to the United States.
4. LSTM models can require significant computational resources when model complexity increases.
5. Bidirectional models may not always be suitable for real-time forecasting when future information is unavailable at prediction time.

6. The study evaluates a relatively limited set of LSTM architectures. These limitations suggest that the model's generalization to other countries and forecasting conditions should be studied further.

12. Critical Analysis

Provide your own analysis of the research article:

Is the selected deep learning architecture appropriate for the problem?

Is the dataset adequate?

Are the evaluation metrics appropriate?

Are the experimental results convincing?

Could another Unit IV technique provide better performance?

Is LSTM appropriate for this problem?

Yes.

COVID-19 case counts form a time series, and LSTM is designed to learn temporal dependencies. Therefore, the selected architecture is appropriate.

Is the dataset adequate?

The dataset is useful because it contains real-world daily COVID-19 observations over a substantial period. However, using only univariate data limits the amount of information available to the model.

Are the evaluation metrics appropriate?

MAPE is appropriate for evaluating forecasting performance because it expresses prediction error as a percentage.

However, additional metrics such as **MAE, RMSE, and R^2** could provide a more comprehensive evaluation.

Are the results convincing?

The comparison between three LSTM architectures provides useful evidence. The reported finding that Bidirectional LSTM performed best is meaningful, but additional datasets and evaluation metrics would make the conclusions stronger.

Could another technique perform better?

Yes. Hybrid models combining LSTM with CNN, attention mechanisms, Transformer models, or other time-series architectures could potentially improve performance.

13. Future Scope

Suggest possible improvements or future research directions based on the limitations identified in the article.

Future research could improve the study in several ways:

1. Include **multivariate data** such as vaccination rates, mobility, hospitalization, weather, and population information.
2. Test the models on datasets from different countries.
3. Compare LSTM with **GRU and Transformer models**.
4. Investigate attention-based LSTM architectures.
5. Use additional evaluation metrics such as MAE and RMSE.
6. Perform longer-term forecasting.
7. Investigate real-time prediction systems.
8. Use ensemble and hybrid deep learning models.
9. Apply explainable AI techniques to understand model predictions.
10. Optimize the models for deployment in healthcare decision-support systems.

14. Conclusion

Provide a brief conclusion summarizing:

The research problem

Proposed model

Major results

Your overall assessment of the article

The reviewed article investigates the use of **LSTM-based deep learning models for predicting univariate COVID-19 time-series data**. The researchers compared Vanilla LSTM, Bidirectional LSTM, and Stacked LSTM architectures using COVID-19 case data collected from the CDC.

The dataset consisted of daily U.S. COVID-19 cases from January 23, 2020 to May 25, 2022, with 743 samples used for training and 104 samples for testing. TensorFlow was used for implementation and KerasTuner was used for hyperparameter optimization.

Among the evaluated models, **Bidirectional LSTM achieved the best accuracy** according to the study.

Overall, the article demonstrates that LSTM-based architectures are useful for sequential time-series prediction. However, future research should incorporate more variables, larger and more diverse datasets, additional evaluation metrics, and newer architectures such as attention-based models and Transformers.

15. References

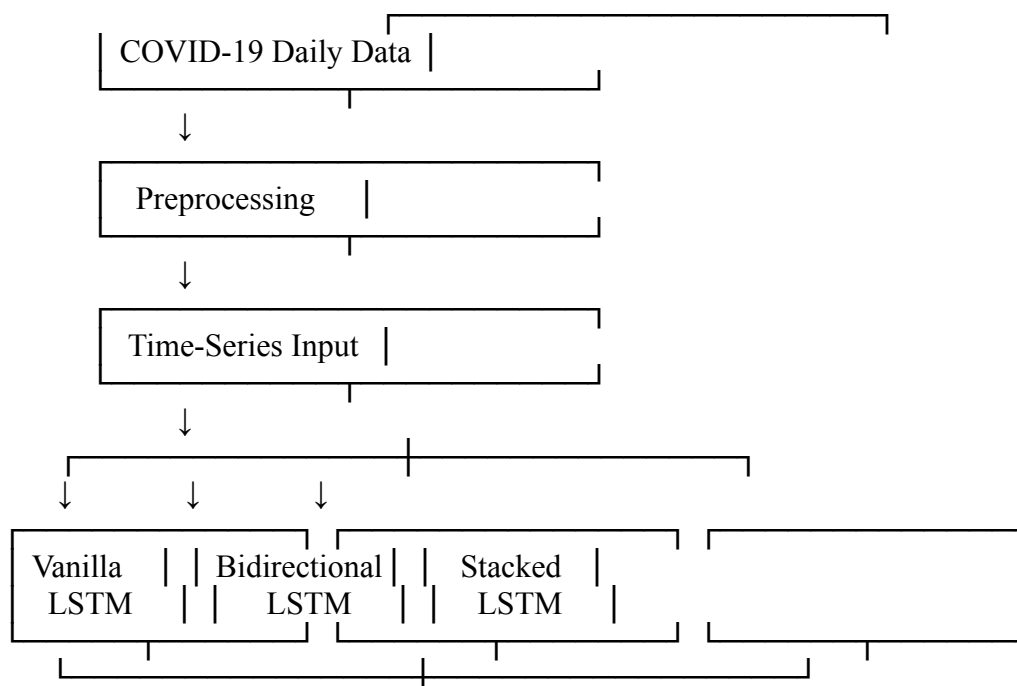
Provide the complete citation of the selected research article and any additional references used in the review.

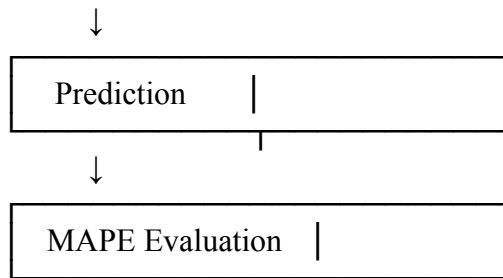
Main Research Article:

Kharel, A., Zarean, Z., & Kaur, D. (2024). *Long Short-Term Memory (LSTM) Based Deep Learning Models for Predicting Univariate Time Series Data*. **International Journal of Machine Learning**, 14(1), 30–37. DOI: 10.18178/ijml.2024.14.1.1154.

Additional Reference:

Ibebuchi, C. C., & Richman, M. B. (2024). *Deep learning with autoencoders and LSTM for ENSO forecasting*. **Climate Dynamics**, 62, 5683–5697. DOI: 10.1007/s00382-024-07180-8.





RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools

Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
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